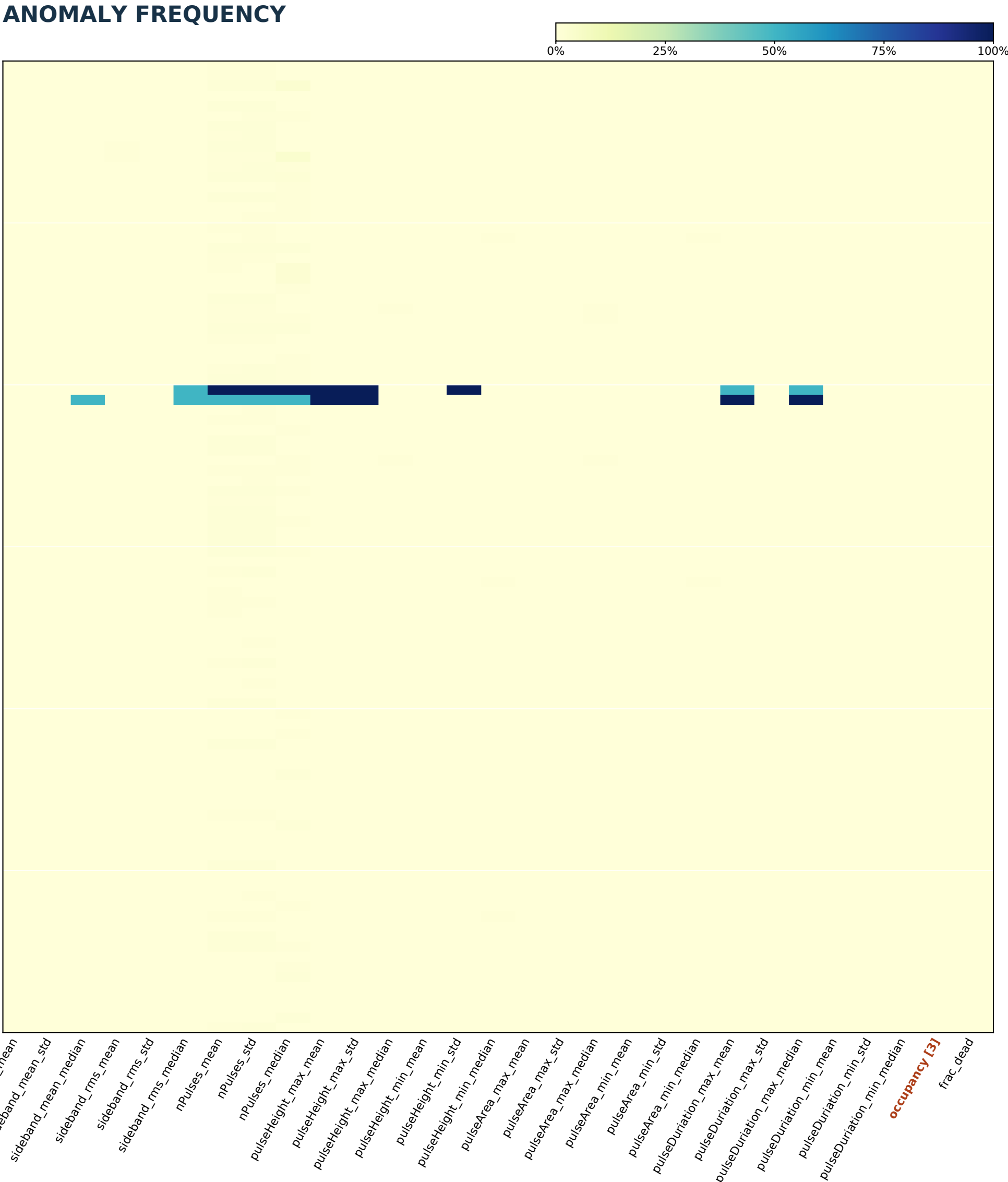
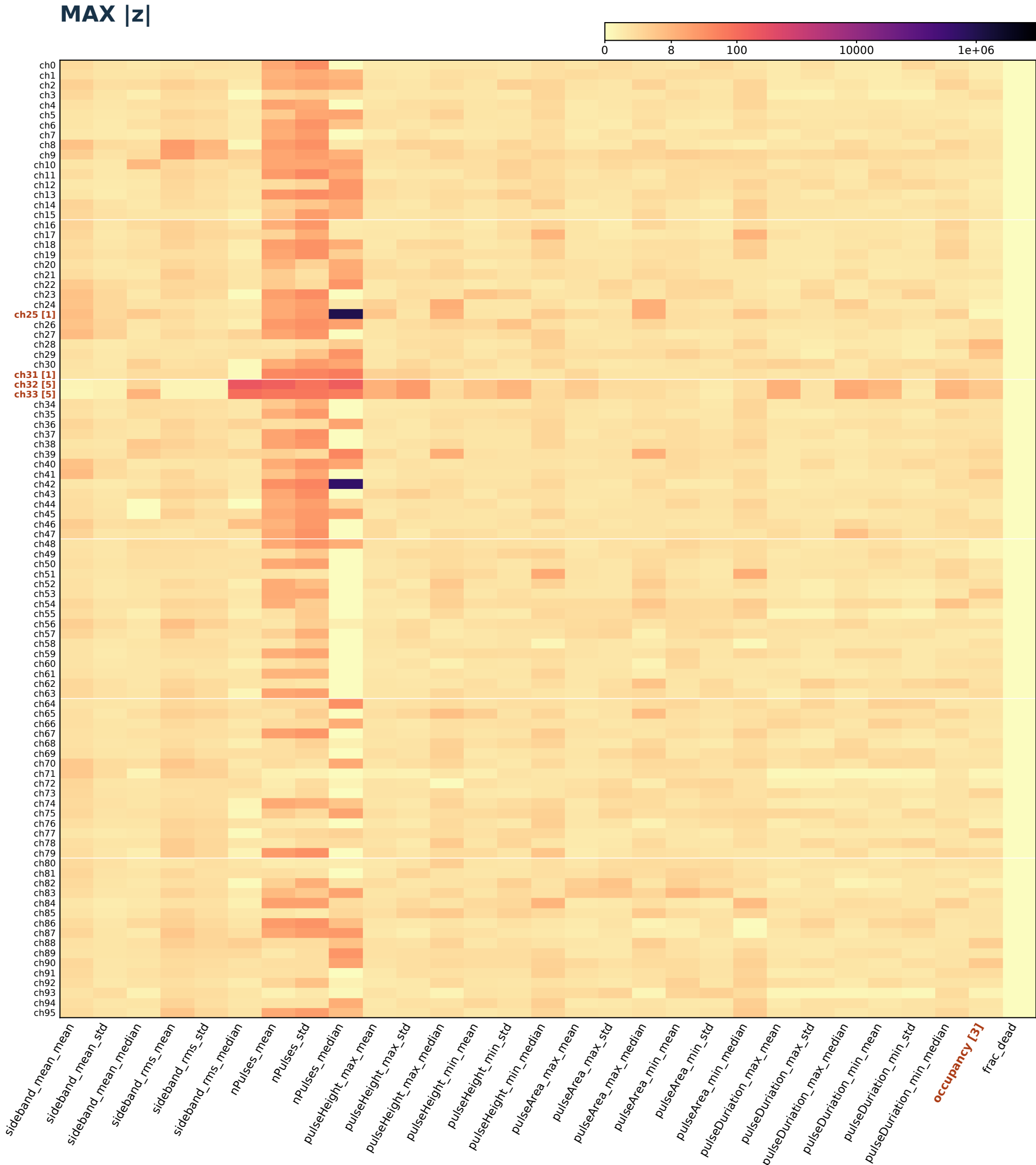


Run 1642 | Observed Signature and LLM-Mentioned Evidence

Frozen IF reference | 96 channels x 29 features | 202/202 valid detector subruns | strict threshold $|z| > 8$ | 5 saved Sol trials

Matrix: digitizer-only model; original feature order and settings. Orange label [n] = explicit text mention in n/5 trials, not internal model attribution.



Max values are NOT clipped; nonlinear color scale resolves both small and million-scale z . Frequency uses each cell's own valid-subrun denominator.

Grey = undefined / masked / no data, not zero. Missing channels are excluded from cell denominators and separately counted below. Orange cell borders require an explicit channel-feature link.

OBSERVED MATRIX

202/202 subruns have defined z values; 0 missing/empty/no- z .

10 cells exceed $|z| > 8$ in at least 80% of their valid subruns.

Most persistent cells (not a cause ranking):

ch32 / nPulses_median: 100.0%; max 179; n=2.

ch32 / nPulses_mean: 100.0%; max 154; n=2.

ch32 / nPulses_std: 100.0%; max 79.9; n=2.

ch32 / pulseHeight_max_std: 100.0%; max 19.5; n=2.

Channel absence is separate from z :

ch32: absent in 200/202 detector subruns.

ch33: absent in 200/202 detector subruns.

LLM-MENTIONED EVIDENCE

Counts are trials, not word occurrences.
Text counts include hypotheses/checks; observed-section mentions are shown separately.
LVDS: 5/5 text; 4/5 observed-section.
LVDS pin16: 5/5 text; 5/5 observed-section.
missing-channel behavior: 5/5 text; 5/5 observed-section.
trigger mask: 5/5 text; 4/5 observed-section.
configuration: 5/5 text; 1/5 observed-section.
trigger rate: 5/5 text; 5/5 observed-section.
board matching: 5/5 text; 0/5 observed-section.
readout failure/dropout: 1/5 text; 0/5 observed-section.
Unambiguous target-observation cell links: 0.
No channel-feature Cartesian expansion. Generic pulse statistics, LVDS pins, mask state and missing-channel text remain notes.

PHASE 1 DIAGNOSES (saved category)

Trial 1: LVDS signal-path dropout affecting channels 32/33

Trial 2: LVDS signal-path loss affecting channels 32/33

Trial 3: persistent_LVDS_signal_loss

Trial 4: persistent LVDS channel dropout

Trial 5: LVDS channel-pair signal loss

Reading across trials: All five categories describe localized LVDS signal loss; the physical root cause remains qualified.

Category text is reproduced from saved responses; it is not a true-cause label.